## 5.1 Conditioning: changing the denominator

**Key point · Definition and results.** For $P(A) > 0$, the probability of B given A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table of counts based on equally likely outcomes, it is the count of $A \cap B$ divided by the count of A. In general $P_A(B) \neq P_B(A)$.

**Status and justification.** Definition. Multiplying by $P(A)$ gives the rule $P(A \cap B) = P(A)P_A(B)$. The condition $P(A) > 0$ is essential.

**Example 1.** Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} = 0.3$, whereas $P_B(A) = \dfrac{12}{30} = 0.4$. $P(A \cap B) = 0.12$.

**Watch out.** "Among the A's" fixes the denominator. A conditional probability is not an intersection.

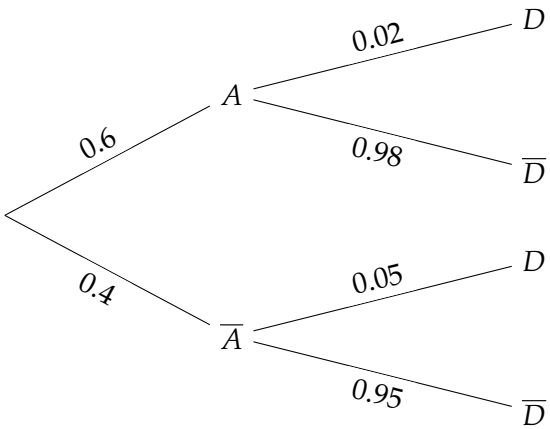
**Application.** Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

## 5.2 Trees, partitions and total probability

**Key point · Definition and results.** A partition $A_1, \ldots, A_k$ is a family of disjoint events whose union is the sample space. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. On a tree, the probabilities leaving the same node sum to 1; the probabilities along a path are multiplied; disjoint paths are combined by adding.

**Status and justification.** $B$ is the disjoint union of the sets $B \cap A_i$. By additivity and by the definition of conditional probability, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

**Example 2.** Two machines supply 60% and 40% of a production; their defect rates are 2% and 5%. $P(D) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.032$. Among the defective items, the first machine accounts for $\dfrac{0.012}{0.032} = \dfrac{3}{8}$.




A denotes the machine that supplies 60 percent of the production and D a defect. For example, $0.02 = P_A(D)$: the second-level probabilities are conditional.

**Watch out.** The simple average of the rates is correct only if the branches have equal weights.

**Application.** $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

## 5.3 Independence and disjointness

**Key point · Definition and results.** A and B are independent if $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. They are disjoint if $A \cap B = \varnothing$. Two disjoint events of strictly positive probability are not independent.

**Status and justification.** The conditional criterion follows from a division by $P(A) > 0$. For disjoint events of positive probability, the left-hand side is 0 and the right-hand product is strictly positive.

**Example 3.** With a fair die, let A: the result is even, B: the result is at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: independent. A and "the result is odd" are disjoint but not independent.

**Watch out.** Independence in a model must be justified or stated explicitly; "two different events" is not enough.

**Application.** Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent?

## 5.4 Successive trials and Bernoulli trials

**Key point · Definition and results.** A Bernoulli trial has two outcomes: success, with probability p, and failure, with probability $1 - p$. In a repetition of n independent, identical trials, with $n \leq 4$ here, the branch probabilities remain p and $1 - p$. A path with k successes has probability $p^k(1-p)^{n-k}$; we count the paths that realise the event.

**Status and justification.** The probability of a path is a product, by independence. The paths corresponding to an event are disjoint: their probabilities add.

**Example 4.** For three trials with $p = \frac{1}{2}$, exactly two successes correspond to SSF, SFS, FSS. Each path has probability $\frac{1}{8}$, so the total is $\frac{3}{8}$. At least one success has probability $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

**Watch out.** When drawing without replacement from an urn, the probabilities generally change after the first draw: the independent model may be wrong.

**Application.** Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

## 5.5 Simulating and discussing the model

**Key point · Definition and results.** A pseudo-random simulation produces frequencies that fluctuate around a probability of the model. For a uniform point in the square $[0,1]^2$, the proportion satisfying $y \leq x^2$ estimates the area under the parabola; the proportion satisfying $x^2 + y^2 \leq 1$, multiplied by 4, estimates $\pi$. Repetitions must use new draws.

**Status and justification.** Simulation principle accepted and explored experimentally. The equality of probability and area is an interpretation of the uniform draw; integration is not a prerequisite. A simulation does not prove an exact value.

**Example 5.** Among 10000 points, if 7840 lie in the quarter disc, the estimate of $\pi$ is $\dfrac{4 \times 7840}{10000} = 3.136$. Two trials of the same size generally give different values.

**Watch out.** Fixing a seed makes a trial reproducible; it neither removes the randomness of the model nor guarantees a specified accuracy.

**Application.** A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate?